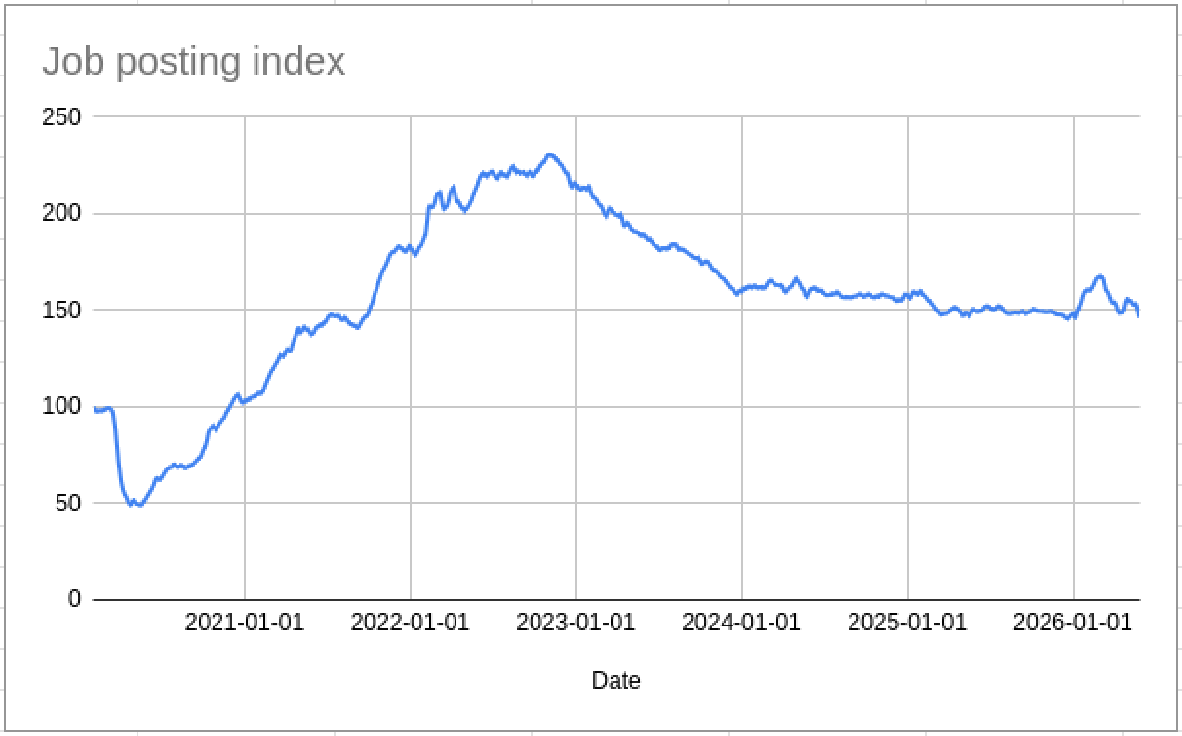
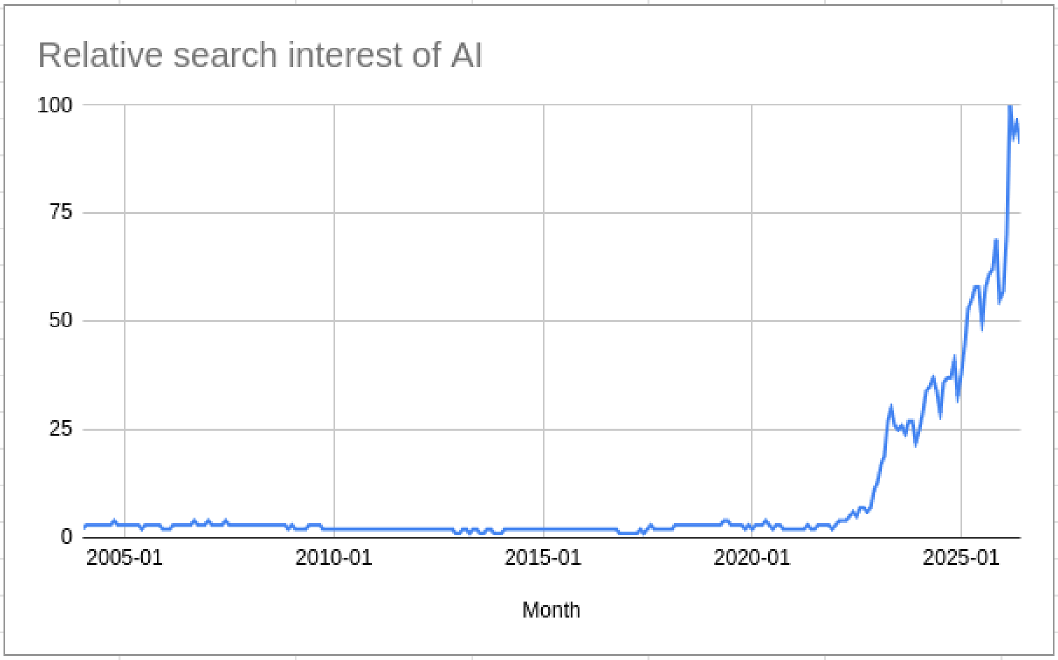
Graph Skew



- Positive skew



- Positive skew

Relationship:

- As interest for AI grew, postings for jobs saw an decrease indicating a correlation with AI growth and job declines.

Logs:

- Existence check: Checked if it exists with zero errors
- Type check: Checked the type of data they were
- Range check: making sure the range is accurate

Security Control	Implementation Status	Justification for Selection
Off-site Backups	[x] Completed (External HDD, Github)	To decrease the risk of physical data loss (fire or hardware failure) by ensuring copies exist in separate locations.

Version Archiving	[x] Completed (Github)	To maintain a history of project development and allow for "rollbacks" if a file becomes corrupted or a logic error is introduced in later stages.
Secure Disposal	[ ] Completed (Final task in project plan)	To uphold data privacy and the Australian Privacy Principle 2 (APP2) by ensuring that survey data is permanently deleted once the research is concluded.
Access Control	[x] Completed (Password protection on MySQL, Shared access protection on google sheets)	To prevent unauthorised access and modifications of the raw job posting and search interest datasets, ensuring data integrity

Test Case	Test Data	Expected Result	Actual Result	Action Taken
Boolean Query	`SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings > 100 AND Date = '2026-04-21';` `SELECT * FROM `Relative search interest of AI -`	Returns specific records matching both criteria.	Returned correct records for April 2026.	None; test passed.

	multiTimeline` WHERE `Relative search interest in Australia(0-100)` > 99 AND Month = '2026-03';			
Range Validation	Inputting SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings = -5.0; SELECT * FROM `Relative search interest of AI - multiTimeline` WHERE `Relative search interest in Australia(0-100)` = -5.0;	Database should return 0 rows as they are beyond the scope of the data.	Database returned 0 rows.	None; test passed.

Test Case	Test Data	Expected	Actual	Action Taken

Pearson's r	`=CORREL(A:A,B:B)`	A value between -1 and 1 showing the relationship.	Returned `0.37` for job posting index and `0.58` for relative search interest of AI	None; test passed.
Mean Calculation	`=AVERAGE(B:B)`	Returns exactly `9.05` for relative search interest of AI and `155.91` for postings	Returned `9.05` for relative search interest of AI and `155.91` for postings	None; test passed.

Test Case	Visual Target	Expected	Actual	Action Taken
Data Accuracy	Job Posting and relative search interest in ai line chart	Peak in late 2022	Line charts peak late 2022	None; test passed
Readability	Blue/Beige Color Scheme	Text must be readable against the background.	Blue text on beige background too light decreasing readability	Changed font color to Navy Blue in order to improve readability

Evaluation Criteria	Measurement Method	Evaluation Finding
Functional Requirements	Check against the original list of findings to be presented.	-
Readability & Design	User testing survey with five university students	-
Data Integrity	Verification of data sources against the final charts.	-

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Future evaluation plan

- **Timeframe:** This solution will be reevaluated 6 months after submission in order to give adequate time for users to develop in depth feedback.

- **Technique:** after 6 months, 10 university students who were willing to participate in evaluating the infographic will give adequate and indepth feedback to my solution.

Functional and non functional aspects of the infographic

Functional:

- Data findings listed
- Data graphs provided

Nonfunctional:

- White space
- Beige and blue color scheme

Problem Solving Methodology:

Analysis:

- Solution requirements: Throughout the making of the infographic I have developed functional requirements such as providing graphs and presenting findings alongside nonfunctional requirements such as a beige and blue design alongside white space
- Solution constraint:Throughout the making of the infographic there has been technological constraints such as exporting the graphs without the white background.
- Scope of solution: Explicit exclusion of mid and senior level workers. Strict focus on newly graduated australians entering the workforce

Design:

- Solution design: Found the template design in which i found visually appealing. Used this design for the official infographic
- Evaluation criteria:

Development:

- Manipulation: Used SQL code in order to demonstrate my validation technique in the infographics documentation
- Validation: Used SQL code and spreadsheet formulas to verify the findings of my infographic
- Testing: Tested loading speed of the infographic as well as testing to see if all information is presented correctly
- Documentation: Documented the process of making the infographic through a comment

Evaluation:

- Strategy: Plan to evaluate in a six month in order to see if it is still effective and efficient
- Report: Reported the

Justification:

Improvements:

- If i were to approach this differently i would improve the design stage of the problem solving methodology. Specifically the solution design. I would have picked different solution designs that looked more visually appealing.

Project Plan Changes justifications:

- Due to criteria 7 requiring extra time, i have made changes to the gantt chart format in order to communicate this information

Evaluation criteria:

1. Functional and non functional requirements:

Functional:

- Solution must present all findings ✔️
- Solution must display data in at least two visual forms✔️

Non functional:

- Solution must be easy to navigate for new workers and graduates ✔️
- Solution must not be messy so it is easy to understand ✔️

2. Data integrity:
    - Data is from popular and reputable sources such as organisations ✔️
    - Data is anonymous, upholding to the APP2 ✔️
    - Data is from the last two years, making it reliable ✔️
3. Readability
    - -Solution must be easy to read and understand
    - Solution layout must be easy to understand ✔️
4. Design
    - Solution must have a visually appealing design ✔️
    - Solutions colour scheme must not take away from the readability of the solution.

✔️

Project Plan Effectiveness:

- The gantt project was an effective tool in order to allocate adequate time in the completion of each part of the assessment criteria
  - The delay in criteria 7 threatened the overall structure of the gantt project however this was fixed through working efficiently through getting more work done in the time i had, ensuring that the rest of the gantt project wont fall into a "domino effect"
- This is reflected through the new version of the gantt project. When compared to the previous one submitted for criteria 1-5, there is a delay for criteria 7 submission reflected onto it.

All SQL codes used:

```
SELECT COUNT(*) FROM `Job posting index - job-postings-headline-index`;

SELECT
   CASE
      WHEN COUNT(Postings) > 100 THEN 'TRUE'
      ELSE 'FALSE'
   END AS Is_High_Volume
FROM `Job posting index - job-postings-headline-index`;

/*Existence checking*/
 SELECT * FROM `Relative search interest of AI - multiTimeline` WHERE `Relative search interest in Australia(0-100)` IS NULL;
 SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings IS NULL;

/*Range checking*/
 SELECT * FROM `Relative search interest of AI - multiTimeline` WHERE `Relative search interest in Australia(0-100)` < 0;
 SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings < 0;

/*Type checking*/
  DESCRIBE `Relative search interest of AI - multiTimeline`;
  DESCRIBE `Job posting index - job-postings-headline-index`;

/*Boolean Query*/
```

```sql
SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings > 100
AND Date = '2026-04-21';
SELECT * FROM `Relative search interest of AI - multiTimeline` WHERE `Relative search
interest in Australia(0-100)` > 99 AND Month = '2026-03';

/*Range Validation*/
SELECT * FROM `Job posting index - job-postings-headline-index` WHERE Postings = -5.0;
SELECT * FROM `Relative search interest of AI - multiTimeline` WHERE `Relative search
interest in Australia(0-100)` = -5.0;
```